# Sagar Gupta

Cloud Consultant – DevOps / MLOps | AWS Professional Services | Hyderabad, India

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## Summary

AI/ML Engineer with two years at AWS Professional Services shipping production ML systems. Trained dense neural network on 244K+ LeetCode contest records to MAE ∼14. Architected medical image classification pipeline on SageMaker with blue-green endpoints, auto-scaling, and drift detection. Built LSTM stock predictor and neural network framework from scratch in NumPy. Upstream contributor to Feast feature store, Apache Airflow, and Prefect. 6x AWS/Terraform certified including **AWS Machine Learning Engineer – Associate** and **AI Practitioner**. LeetCode Knight, 2007 peak.

## Experience

**Amazon Web Services (AWS)** Aug 2024 – Present
*Cloud Consultant – Professional Services (DevOps/MLOps)* Hyderabad, India

- **MLOps Pipeline – SageMaker (SME Program, Jun-Aug 2025):** Trained and shipped medical image classification model on SageMaker with blue-green endpoint deployment, auto-scaling, and Model Monitor drift detection – zero-downtime updates with automatic rollback. Full IaC via Terraform + CodePipeline; API Gateway + Lambda inference API behind CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (∼90%) building the AWS foundation for a data-centre exit and a second landing zone entirely in Terraform: Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines. Delivered two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect, plus self-service network onboarding. Used Kiro and Claude Code for requirement analysis, specs, and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization. Project closed with a **10/10 CSAT**.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns on high-sensitivity financial workloads. Used Amazon Q to accelerate code analysis and migration assessments.
- **Internal Impact:** Published a second AWS sample, Terraform for AWS Organizations governance on Control Tower (aws-samples, MIT-0), covering all six policy types, Control Tower controls, and delegated admin for 15 services. MLOps SME Program Graduate (2025). Published APG pattern *Secure PR Previews and Approval-Gated Releases for Amplify Gen 2* (plus 2 more peer-reviewed in publication pipeline). 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Associate Speaker Certified.

**Amazon Web Services (AWS)** Mar 2024 – Jul 2024
*Associate Professional Services – DevOps (Internship)* Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Established observability with X-Ray and CloudWatch distributed tracing.

**Ikarus-3D** May 2023 – Jul 2023
*Software Developer Associate (Internship)* Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code quality analysis into the delivery pipeline.

## Industry Certifications (6)

**AWS**: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
**AWS**: AI Practitioner | Cloud Practitioner **HashiCorp**: Terraform Associate (003)

## Education

**National Institute of Technology Warangal, Telangana** Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

**Sanskar College of Professional Studies (DAVV University), Indore** May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

## Projects

**LeetCode Rating Predictor** Dec 2023
*Python, TensorFlow, FastAPI, React, GraphQL* GitHub | Live
◇ Trained dense neural network on **244K+ contest records** reaching **MAE ∼14** for post-contest rating predictions; feature engineering pipeline pulls raw contest data via LeetCode GraphQL API.
◇ Served the TensorFlow model behind a FastAPI backend with a React frontend for real-time user predictions.

**GroundZeroNN** Oct 2023
*Python, NumPy, Neural Networks from Scratch* GitHub
◇ Built a neural network framework in pure Python/NumPy – forward/backward propagation, gradient descent, activation functions, and batch normalization implemented from scratch without any ML library.

**StockSage AI** Nov 2023
*Python, TensorFlow, LSTM, Streamlit* GitHub
◇ LSTM stock price predictor combining technical indicators (RSI, MACD, Bollinger Bands) with sequential price data; interactive Streamlit dashboard for real-time predictions and historical back-tests.

**Bedrock Multi-Model MCP Server** 2026
*Python, Amazon Bedrock, MCP Protocol* GitHub
◇ MCP server routing tool calls across 500+ Amazon Bedrock foundation models through a single endpoint – Claude, Llama, Mistral, Titan – with per-model streaming and cost telemetry.

**Claude Cost Optimizer** 2026
*TypeScript, Anthropic API, React* Live
◇ Analyzes Claude usage traces to recommend model + prompt-caching strategies that cut spend 30–60% without accuracy loss.

*Other Notable Projects:* GitScope (Chrome extension, GitHub analytics), Ledger Sync (personal-finance reconciler), Instagram Likes Leaderboard.

## Open Source Contributions

*(7 merged upstream, 1 open)*

- **axios** (109K☆) – Synthesize AxiosError message from AggregateError on dual-stack connection failures (#11059, merged)
- **apache/airflow** (44K☆) – Add `template_fields` to SalesforceBulkOperator (#63109, merged)
- **PrefectHQ/prefect** (21K☆) – Document custom deployment steps as Python functions (#20956, merged)
- **feast-dev/feast** (9K☆) – Claude Code agent skills for Feast feature store (#6081, merged)
- **awslabs/agent-plugins** – Guard defusedxml extension load (#132, merged)
- **lucide** (24K☆) – Vue Icon kebab-case icon-node prop fix (#4512, merged)
- **aws-ia/terraform-aws-vpc** – Cloud WAN attachment-replacement root cause; fix co-authored into v4.8.0 (merged)
- **hashicorp/terraform-provider-aws** – IoT CloudWatch metric substitution validation (#46867, open)

## Technical Skills

**ML & Deep Learning**: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas; Dense Networks, CNNs, LSTMs, Ensemble Methods, Feature Engineering, Model Evaluation (MAE/RMSE, precision/recall)
**MLOps on AWS**: SageMaker (Training, Pipelines, Endpoints, Model Monitor, Drift Detection), Blue-Green Deployments, Model Versioning, CodePipeline, Feast Feature Store
**GenAI & LLMs**: Amazon Bedrock, RAG Pipelines, LLM APIs (Anthropic, OpenAI), MCP Protocol, Prompt Engineering, Claude Code, Amazon Q, Kiro
**Serving & Infrastructure**: FastAPI, Lambda, API Gateway, CloudFront, Docker, Terraform, GitHub Actions, GraphQL
**Languages**: Python, SQL, TypeScript, JavaScript, HCL, C++, Java

## Achievements

- **MLOps SME Program Graduate** (AWS, 2025). Published APG pattern: *Secure PR Previews and Approval-Gated Releases for Amplify Gen*
- **LeetCode Knight** – 2007 peak rating, 1200+ problems solved, 100+ contests. **GeeksforGeeks** 400+ DSA problems. **HackerRank** 6-Star Pr
- **Rank 1289 (Top 5%)** – Google Kick Start Round E '22. *Certificate*
- **1st Place** – Coding Cauldron '23 (NIT Trichy); QFiesta IGNOSI '22 (NIT Warangal).